

# APPENDIX II — NEU5GC SEVERITY INDEX

## *Hidden Dietary Sources of Bovine Neu5Gc, Ammonium Catalysts, and Associated Trigger Compounds*

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### 1. Purpose and Scope

This index catalogues 148 specific ingredients, food additives, processing agents, and product categories representing known or probable sources of dietary N-glycolylneuraminic acid (Neu5Gc), ammonium-based inflammatory catalysts, and associated trigger compounds within the Sialic-Glycan Inflammatory Cascade (SGIC) framework. It is the detection backbone of the Acnerase food-label scanning application and the evidentiary foundation for the severity claims in every white paper in the Hamer Glycan Systems series.

The index exists because the primary challenge in avoiding dietary Neu5Gc is hidden sources: bovine gelatin in pharmaceutical capsule shells, anhydrous milk fat in confectionery glazes, caramel colour in beverages, natural flavour systems in nominally vegetable-based foods built on bovine broth foundations. These hidden sources are why elimination protocols fail, and why Acnerase was designed to identify ingredients before consumption rather than relying on label literacy alone.

Severity ratings are derived from 23 years of systematic self-provocation and elimination trials (1981–2026), each ingredient category tested separately and repeatedly, outcomes documented by lesion characteristics, timing, severity, and resolution. Baseline: whole cow's milk = Severity 1, Resolution A.

### 2. Rating System

**Severity Index 1–5 — calibrated against whole cow's milk as Severity 1 baseline:**

| Rating          | Descriptor        | Typical Lesion Profile                         | Resolution                |
|-----------------|-------------------|------------------------------------------------|---------------------------|
| 1 — Mild        | MILD              | Small, transient; milky; no odour; rapid onset | A — Short-lived           |
| 2 — Moderate    | MODERATE          | Larger; persistent; may develop odour          | B — 24–72 hours           |
| 3 — Mod. Severe | MODERATELY SEVERE | Deep-seated; gram-negative odour               | C — Up to one week        |
| 4 — Severe      | SEVERE            | Large nodular; deep; slow resolution           | D — Greater than one week |
| 5 — Suppurative | SUPPURATIVE       | Cystic; painful; weeks to resolve              | E — Weeks                 |

*Ratings reflect the maximum-sensitivity purge state. In the chronically loaded state, lower-severity exposures may be attenuated. Upon successful elimination, re-exposure sensitivity amplifies significantly. Manufacturer variation applies.*

### 3. The 148-Ingredient Neu5Gc Severity Index

Ingredients listed alphabetically. Colour coding in the Severity column reflects severity tier. The Acnerase application extends each category with 150+ label synonyms.

| # | Ingredient / Label Term | Severity | Description | Source Notes |
|---|-------------------------|----------|-------------|--------------|
|---|-------------------------|----------|-------------|--------------|

|    |                                                      |                        |                                                                                                                                                                                                                     |                                      |
|----|------------------------------------------------------|------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------|
| 1  | <b>Acetylated Monoglycerides (ACETEM / AMG)</b>      | <b>2 — Moderate</b>    | Non-ionic emulsifiers used in food to improve texture, stability, and shelf life. They bridge the oil and water phases, making them effective in dairy-containing emulsions, baked goods, and whipped products.     | Occasionally derived from cows       |
| 2  | <b>Acid Whey</b>                                     | <b>2 — Moderate</b>    | Liquid byproduct of acid-coagulated cheese (e.g., cottage cheese, ricotta, Greek yogurt). Used in protein powders, baked goods, and animal feed.                                                                    | Always dairy                         |
| 3  | <b>Acidophilus</b>                                   | <b>2 — Moderate</b>    | Lactobacillus acidophilus, a probiotic bacterium cultured on milk to ferment lactose. Found in yogurt, kefir, supplements, and probiotic dairy drinks.                                                              | Cultured on dairy media              |
| 4  | <b>Albumen</b>                                       | <b>2 — Moderate</b>    | Protein fraction historically referring to egg white but in dairy contexts denotes milk albumin. Used as a binder, foaming agent, and protein supplement.                                                           | Dairy or egg source                  |
| 5  | <b>Albumin</b>                                       | <b>2 — Moderate</b>    | Water-soluble protein found in milk (lactalbumin), eggs, and blood. Used in protein supplements, bakery products, and pharmaceutical excipients.                                                                    | Dairy or animal source               |
| 6  | <b>Alcohol (general)</b>                             | <b>4 — Severe</b>      | Fermented beverages and ethanol used in flavorings, extracts, and cooking. Production often uses dairy-derived clarifiers (casein, isinglass) and yeast nutrients containing ammonium compounds.                    | May involve dairy fining agents      |
| 7  | <b>Ammonium</b>                                      | <b>5 — Suppurative</b> | Ammonium-based compounds (ammonium chloride, bicarbonate, sulfate, hydroxide) used as leavening agents, dough conditioners, pH regulators, and processing aids. Strong inflammatory trigger at high concentrations. | Concentrated ammonium                |
| 8  | <b>Ammonium Caseinate</b>                            | <b>5 — Suppurative</b> | Casein neutralized with ammonium hydroxide. Used as an emulsifier, whitener, and protein source in coffee creamers, processed cheese, and meat products.                                                            | Always dairy + concentrated ammonium |
| 9  | <b>Anhydrous Milk Fat (AMF)</b>                      | <b>3 — Mod. Severe</b> | Concentrated milk fat (>99.8% fat) produced by removing water and non-fat solids from butter or cream. Used in confectionery, bakery, recombined milk, and ice cream.                                               | Always dairy, butter-tier            |
| 10 | <b>Artificial Butter / Artificial Butter Flavour</b> | <b>4 — Severe</b>      | Synthetic flavor compounds (notably diacetyl and acetoin) formulated to mimic butter aroma in margarine, popcorn topping, baked goods, and sprays. Often delivered in oil or maltodextrin carriers.                 | Synthetic; may contain dairy         |
| 11 | <b>Artificial Colour / Artificial Color</b>          | <b>4 — Severe</b>      | Synthetic dyes (FD&C colors, azo dyes, lakes) added to processed foods, drugs, and cosmetics. Frequently dispersed in carriers that contain ammonium compounds.                                                     | Ammonium incorporated                |
| 12 | <b>Artificial Flavour / Artificial Flavor</b>        | <b>4 — Severe</b>      | Synthetic flavor compounds blended in solvent or carrier systems (often containing ammonium-based reaction products) to replicate natural tastes.                                                                   | Ammonium incorporated                |
| 13 | <b>Aspic</b>                                         | <b>4 — Severe</b>      | Savory gelatin made from clarified meat or bone stock, used to glaze cold cuts, terrines, and pates. Concentrated animal collagen with curing additives.                                                            | Animal-derived; processed meat       |
| 14 | <b>Beef</b>                                          | <b>4 — Severe</b>      | Bovine muscle meat. Inherently rich in Neu5Gc; severity rises further when cured, smoked, or processed with nitrates and additives.                                                                                 | Bovine, primary Neu5Gc source        |
| 15 | <b>Beer</b>                                          | <b>4 — Severe</b>      | Fermented malt beverage. Often clarified with casein/isinglass and produced with ammonium-containing yeast nutrients; flavored varieties may contain caramel.                                                       | Possible dairy fining                |
| 16 | <b>Beta Lactoglobulin</b>                            | <b>2 — Moderate</b>    | The principal whey protein in cow milk. Used in infant formulas, sports nutrition, and protein-fortified foods.                                                                                                     | Always dairy                         |
| 17 | <b>Binders</b>                                       | <b>4 — Severe</b>      | Generic term for dairy- or meat-derived proteins (casein, whey, gelatin, blood plasma) used to hold processed meats and reformed products together.                                                                 | Often dairy or animal-derived        |
| 18 | <b>Bologna</b>                                       | <b>4 — Severe</b>      | Emulsified cured sausage made from beef and/or pork with nitrates, milk powders, and binders.                                                                                                                       | Processed meat                       |
| 19 | <b>Butter</b>                                        | <b>3 — Mod. Severe</b> | Churned dairy fat (>=80% milk fat). The user's reference baseline for Severity 3 (Moderately Severe).                                                                                                               | Always dairy, baseline trigger       |
| 20 | <b>Butter Extract</b>                                | <b>3 — Mod. Severe</b> | Concentrated butter flavor in oil or alcohol carrier, used in baking and confectionery.                                                                                                                             | Always dairy                         |
| 21 | <b>Butter Fat</b>                                    | <b>3 — Mod. Severe</b> | Lipid fraction of milk separated as butter or anhydrous milk fat. Used in chocolate, ice cream, and bakery.                                                                                                         | Always dairy                         |
| 22 | <b>Butter Flavoured Oil / Butter Flavored Oil</b>    | <b>4 — Severe</b>      | Vegetable oil infused with butter flavoring compounds (often diacetyl). Used as a popcorn topping and bakery release oil.                                                                                           | Flavour additive on oil base         |

|    |                                               |                        |                                                                                                                                                                           |                                |
|----|-----------------------------------------------|------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------|
| 23 | <b>Butter Oil</b>                             | <b>3 — Mod. Severe</b> | Anhydrous butter fat used in chocolate manufacture, recombined dairy, and bakery applications.                                                                            | Always dairy                   |
| 24 | <b>Butter Solids</b>                          | <b>3 — Mod. Severe</b> | Dehydrated butter delivered as a powder or paste for industrial baking and food manufacturing.                                                                            | Always dairy                   |
| 25 | <b>Buttermilk</b>                             | <b>2 — Moderate</b>    | Cultured low-fat milk byproduct of butter making, or fermented skim milk. Used in baking, dressings, and pancakes.                                                        | Always dairy                   |
| 26 | <b>Buttermilk Blend / Powder / Solids</b>     | <b>3 — Mod. Severe</b> | Spray-dried buttermilk used as a flavoring, browning agent, and dairy solid in mixes, baked goods, and confectionery.                                                     | Always dairy, concentrated     |
| 27 | <b>Calcium (added)</b>                        | <b>2 — Moderate</b>    | Mineral added to fortified foods, frequently in the form of calcium caseinate, calcium lactate, or calcium phosphate from dairy origin.                                   | Often dairy-sourced            |
| 28 | <b>Calcium Caseinate</b>                      | <b>2 — Moderate</b>    | Casein neutralized with calcium hydroxide. Used as a whitener, emulsifier, and protein in nutrition bars, coffee whiteners, and processed cheese.                         | Always dairy                   |
| 29 | <b>Capsules (Bovine Gelatin)</b>              | <b>4 — Severe</b>      | Hard and soft pharmaceutical capsules made from bovine hide and bone gelatin.                                                                                             | Bovine, processed              |
| 30 | <b>Caramel</b>                                | <b>5 — Suppurative</b> | Heat-treated sugar product. Class III (ammonia-process) and Class IV (ammonia-sulfite-process) caramel colors are produced with ammonium compounds at high concentration. | Concentrated ammonium          |
| 31 | <b>Caramel Colour / Color (all classes)</b>   | <b>5 — Suppurative</b> | Most widely used food colorant; Classes III and IV are produced by reacting carbohydrates with ammonia or ammonia + sulfite.                                              | Concentrated ammonium          |
| 32 | <b>Caramel Flavour / Flavor</b>               | <b>5 — Suppurative</b> | Caramel-based flavoring blends combining caramelized sugar with ammoniated reaction products.                                                                             | Concentrated ammonium          |
| 33 | <b>Casein</b>                                 | <b>2 — Moderate</b>    | Predominant protein of cow's milk (~80% of milk protein). Used as a binder, emulsifier, and protein source in processed foods.                                            | Always dairy                   |
| 34 | <b>Casein Hydrolysate / Hydrolyzed Casein</b> | <b>2 — Moderate</b>    | Enzymatically broken-down casein peptides used in hypoallergenic formulas, sports nutrition, and clinical foods.                                                          | Always dairy                   |
| 35 | <b>Caseinate (general)</b>                    | <b>2 — Moderate</b>    | Casein neutralized with sodium, calcium, potassium, ammonium, magnesium, or iron. Used as an emulsifier and protein in many processed foods.                              | Always dairy                   |
| 36 | <b>Cheese</b>                                 | <b>2 — Moderate</b>    | Curdled milk product. Severity may increase with processed/aged varieties or added flavor systems.                                                                        | Always dairy                   |
| 37 | <b>Cheese Flavour / Flavor</b>                | <b>4 — Severe</b>      | Concentrated cheese flavoring blends made from enzyme-modified cheese, dairy solids, and reaction flavors.                                                                | Dairy + flavour additive       |
| 38 | <b>Cheese Food</b>                            | <b>2 — Moderate</b>    | Pasteurized blend of natural cheese with whey, milk, emulsifiers, and stabilizers, formulated for melt and slice consistency.                                             | Always dairy, processed        |
| 39 | <b>Chocolate</b>                              | <b>3 — Mod. Severe</b> | Confectionery made from cacao with sugar; milk chocolate adds milk solids and butter fat. Includes lecithin emulsifiers.                                                  | Usually dairy                  |
| 40 | <b>Chocolate Flavour / Flavor</b>             | <b>4 — Severe</b>      | Chocolate-flavor systems for beverages, baked goods, and confectionery, often delivered in dairy carriers.                                                                | Often dairy + flavour additive |
| 41 | <b>Cochineal (Carmine / Red Dye)</b>          | <b>4 — Severe</b>      | Red colorant derived from cochineal insects, typically dispersed in ammonium-containing carrier solutions.                                                                | Insect-derived colorant        |
| 42 | <b>Color / Colour (general)</b>               | <b>4 — Severe</b>      | Catch-all label term covering both natural and synthetic colorants, frequently formulated with ammonium-containing solvents.                                              | Ammonium incorporated          |
| 43 | <b>Coloring / Colouring (natural)</b>         | <b>4 — Severe</b>      | Pigments extracted from plant, animal, or insect sources. Carriers and stabilizers commonly contain ammonium compounds.                                                   | Ammonium incorporated          |
| 44 | <b>Condensed Milk</b>                         | <b>1 — Mild</b>        | Cow's milk concentrated by evaporation, often with added sugar (sweetened condensed milk). Used in desserts and beverages.                                                | Always dairy                   |
| 45 | <b>Cottage Cheese</b>                         | <b>2 — Moderate</b>    | Fresh, mild-flavored curd cheese drained from acid-coagulated milk. May contain cream dressing.                                                                           | Always dairy                   |
| 46 | <b>Cream</b>                                  | <b>2 — Moderate</b>    | Higher-fat fraction skimmed from milk. Used in dairy products, sauces, soups, and desserts.                                                                               | Always dairy                   |
| 47 | <b>Cream Cheese</b>                           | <b>2 — Moderate</b>    | Soft, mild fresh cheese with high butterfat (>=33%). Used in spreads, dips, and baking.                                                                                   | Always dairy                   |
| 48 | <b>Creamers (dairy and non-dairy)</b>         | <b>3 — Mod. Severe</b> | Liquid or powdered coffee whiteners. 'Non-dairy' creamers commonly contain sodium caseinate and butter solids.                                                            | Often dairy                    |
| 49 | <b>Cultured Milk</b>                          | <b>1 — Mild</b>        | Milk fermented with lactic acid bacteria (e.g., buttermilk, kefir base). Used in baking and beverages.                                                                    | Always dairy                   |
| 50 | <b>Curd</b>                                   | <b>2 — Moderate</b>    | Coagulated milk solids separated from whey, used as the base for cheese and paneer.                                                                                       | Always dairy                   |

|    |                                                                  |                        |                                                                                                                                                                       |                              |
|----|------------------------------------------------------------------|------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| 51 | <b>Curd Whey</b>                                                 | <b>2 — Moderate</b>    | Sweet whey collected from rennet-set cheese curds.                                                                                                                    | Always dairy                 |
| 52 | <b>Custard</b>                                                   | <b>3 — Mod. Severe</b> | Cooked mixture of milk or cream with eggs and sugar, used in desserts and bakery fillings.                                                                            | Always dairy                 |
| 53 | <b>Dairy Binders</b>                                             | <b>2 — Moderate</b>    | Functional dairy proteins (casein, whey, milk powder) used to bind processed meats, baked goods, and reformed products.                                               | Always dairy                 |
| 54 | <b>Dairy Butter</b>                                              | <b>3 — Mod. Severe</b> | Industry term distinguishing real butter from artificial 'butter' products.                                                                                           | Always dairy                 |
| 55 | <b>Dairy Products / Dairy Product Solids</b>                     | <b>2 — Moderate</b>    | Generic ingredient labels covering any milk-derived component on a label.                                                                                             | Always dairy                 |
| 56 | <b>Delactosed Whey</b>                                           | <b>2 — Moderate</b>    | Whey with most of its lactose removed; used as a high-protein, low-sugar dairy solid in baked goods and processed foods.                                              | Always dairy                 |
| 57 | <b>Demineralized Whey</b>                                        | <b>2 — Moderate</b>    | Whey with minerals removed via ion exchange or electrodialysis, used in infant formula and confectionery.                                                             | Always dairy                 |
| 58 | <b>Derivative Milk / Milk Derivatives</b>                        | <b>1 — Mild</b>        | Generic label term covering any compound chemically obtained from milk components.                                                                                    | Always dairy                 |
| 59 | <b>Diacetyl</b>                                                  | <b>2 — Moderate</b>    | 2,3-butanedione, a natural and synthetic flavor compound responsible for buttery aroma. Found in butter, cheese, beer, and butter flavorings.                         | Dairy fermentation byproduct |
| 60 | <b>DMS (Dried Milk Solids)</b>                                   | <b>1 — Mild</b>        | Industry abbreviation for dried milk solids used as a generic dairy ingredient in baking and confectionery.                                                           | Always dairy                 |
| 61 | <b>Dried Milk / Dry Milk / Dry Milk Powder / Dry Milk Solids</b> | <b>1 — Mild</b>        | Spray-dried whole or skim cow's milk used as a shelf-stable dairy ingredient in baking, confectionery, and recombined milk.                                           | Always dairy                 |
| 62 | <b>E270 / Lactic Acid</b>                                        | <b>2 — Moderate</b>    | Organic acid produced by bacterial fermentation of carbohydrates (often lactose or whey). Used as an acidulant, preservative, and flavor adjunct.                     | Often dairy-fermented        |
| 63 | <b>Enriched Flour</b>                                            | <b>4 — Severe</b>      | Refined flour fortified with vitamins and iron; commercial enrichment blends often include ammonium chloride/sulfate as dough conditioners and yeast nutrients.       | Ammonium incorporated        |
| 64 | <b>Evaporated Milk</b>                                           | <b>1 — Mild</b>        | Cow's milk concentrated to about 60% of its original volume by evaporation; canned and homogenized.                                                                   | Always dairy                 |
| 65 | <b>Fat Free Milk / Skim Milk</b>                                 | <b>1 — Mild</b>        | Cow's milk with butterfat removed (<0.5%).                                                                                                                            | Always dairy                 |
| 66 | <b>Fat Replacer</b>                                              | <b>4 — Severe</b>      | Carbohydrate-, protein-, or lipid-based ingredient used to replace dietary fat (e.g., Simplex, Olestra). Protein-based replacers are typically whey- or milk-derived. | Often dairy-derived          |
| 67 | <b>Flavor / Flavour (general label term)</b>                     | <b>4 — Severe</b>      | Single-word label term referring to any flavoring substance; lower-concentration ammonium incorporation than 'flavoring/flavouring' formulations.                     | Ammonium incorporated        |
| 68 | <b>Flavored Drinks / Flavoured Drinks</b>                        | <b>4 — Severe</b>      | Beverages containing added flavor systems and colorants; many use ammoniated caramel and complex flavoring blends.                                                    | Ammonium incorporated        |
| 69 | <b>Flavoring / Flavouring</b>                                    | <b>5 — Suppurative</b> | Concentrated flavor systems (often '-ing' gerund usage on labels) carrying high ammonium-reaction-product content.                                                    | Concentrated ammonium        |
| 70 | <b>Fully Cream Milk Powder</b>                                   | <b>1 — Mild</b>        | Whole milk powder retaining all milk fat; used in beverages, confectionery, and recombined milk.                                                                      | Always dairy                 |
| 71 | <b>Galactose</b>                                                 | <b>2 — Moderate</b>    | Monosaccharide derived from the hydrolysis of lactose. Used as a sweetener and metabolic substrate.                                                                   | Dairy-derived                |
| 72 | <b>Gelatin / Gelatine</b>                                        | <b>4 — Severe</b>      | Hydrolyzed collagen from bovine or porcine hide and bones. Used as a gelling, thickening, and stabilizing agent.                                                      | Animal-derived               |
| 73 | <b>Ghee</b>                                                      | <b>3 — Mod. Severe</b> | Clarified butter with milk solids removed by cooking. Stable, high-fat dairy ingredient used in South Asian cooking.                                                  | Always dairy, butter-tier    |
| 74 | <b>Goat Cheese</b>                                               | <b>2 — Moderate</b>    | Cheese made from goat's milk; carries the same Neu5Gc concerns as cow's cheese.                                                                                       | Always dairy                 |
| 75 | <b>Goat Milk</b>                                                 | <b>1 — Mild</b>        | Milk from domestic goats; cross-reactive with cow milk for Neu5Gc-sensitive individuals.                                                                              | Always dairy                 |
| 76 | <b>Half &amp; Half</b>                                           | <b>1 — Mild</b>        | Equal parts whole milk and cream (~10-12% fat).                                                                                                                       | Always dairy                 |
| 77 | <b>High Protein Flour</b>                                        | <b>4 — Severe</b>      | Flour blends boosted with vital wheat gluten, whey protein, or milk protein isolates.                                                                                 | Often dairy-fortified        |
| 78 | <b>Hot Chocolate Mixes</b>                                       | <b>3 — Mod. Severe</b> | Cocoa-based beverage mixes containing milk powder, sugar, and emulsifiers; often include caramel notes.                                                               | Always dairy                 |
| 79 | <b>Hot Dogs</b>                                                  | <b>4 — Severe</b>      | Emulsified cured sausage with binders, nitrites, milk powders, and flavorings.                                                                                        | Processed meat               |

|     |                                                      |                        |                                                                                                                            |                                 |
|-----|------------------------------------------------------|------------------------|----------------------------------------------------------------------------------------------------------------------------|---------------------------------|
| 80  | <b>Hydrolyzed Milk Protein</b>                       | <b>2 — Moderate</b>    | Milk protein broken into smaller peptides by enzymes or acids. Used in hypoallergenic formulas and clinical nutrition.     | Always dairy                    |
| 81  | <b>Hydrolyzed Whey</b>                               | <b>2 — Moderate</b>    | Whey protein broken into peptides. Common in sports nutrition and infant formulas.                                         | Always dairy                    |
| 82  | <b>Ice Cream</b>                                     | <b>3 — Mod. Severe</b> | Frozen dessert based on cream, milk, and sugar; often includes egg yolk, stabilizers, and flavorings.                      | Always dairy                    |
| 83  | <b>Ice Milk</b>                                      | <b>3 — Mod. Severe</b> | Frozen dessert similar to ice cream but with reduced milk fat (2-7%).                                                      | Always dairy                    |
| 84  | <b>Imitation Cheese</b>                              | <b>2 — Moderate</b>    | Cheese-substitute products that frequently retain casein or whey while replacing milk fat with vegetable oil.              | Often contains dairy            |
| 85  | <b>Imitation Sour Cream</b>                          | <b>2 — Moderate</b>    | Sour-cream substitute commonly containing sodium caseinate and dairy solids despite the 'imitation' label.                 | Often contains dairy            |
| 86  | <b>Iron Caseinate</b>                                | <b>2 — Moderate</b>    | Iron-fortified casein used in nutritional and clinical foods.                                                              | Always dairy                    |
| 87  | <b>Kefir</b>                                         | <b>2 — Moderate</b>    | Fermented milk drink made with kefir grains (yeasts and lactic acid bacteria).                                             | Always dairy                    |
| 88  | <b>Kourmiss / Koumiss</b>                            | <b>2 — Moderate</b>    | Fermented mare's or cow's milk beverage of Central Asian origin.                                                           | Always dairy                    |
| 89  | <b>Lactaglobulin</b>                                 | <b>2 — Moderate</b>    | Synonym for beta-lactoglobulin, the dominant whey protein in cow milk.                                                     | Always dairy                    |
| 90  | <b>Lactaid Milk</b>                                  | <b>1 — Mild</b>        | Cow's milk pretreated with lactase enzyme to break down lactose; Neu5Gc and milk proteins remain unchanged.                | Always dairy                    |
| 91  | <b>Lactalbumin</b>                                   | <b>2 — Moderate</b>    | Whey protein fraction (alpha-lactalbumin) used in infant formulas and supplements.                                         | Always dairy                    |
| 92  | <b>Lactalbumin Phosphate</b>                         | <b>2 — Moderate</b>    | Phosphate-modified whey protein used as a dough conditioner and protein source.                                            | Always dairy                    |
| 93  | <b>Lactate / Lactate Solids</b>                      | <b>2 — Moderate</b>    | Salts of lactic acid (sodium, calcium, potassium, magnesium lactate) used as preservatives, humectants, and pH regulators. | Often dairy-fermented           |
| 94  | <b>Lactitol Monohydrate</b>                          | <b>2 — Moderate</b>    | Sugar alcohol derived from lactose by hydrogenation. Used as a low-calorie sweetener and laxative.                         | Always dairy-derived            |
| 95  | <b>Lactoferrin</b>                                   | <b>2 — Moderate</b>    | Iron-binding glycoprotein found in milk; isolated from whey for use in infant formulas and supplements.                    | Always dairy                    |
| 96  | <b>Lactose</b>                                       | <b>1 — Mild</b>        | Milk sugar (disaccharide of glucose + galactose). Used as a filler, sweetener, and tablet excipient.                       | Always dairy                    |
| 97  | <b>Lactose Free Milk / Lactulose Free Milk</b>       | <b>1 — Mild</b>        | Cow's milk with lactose enzymatically split into glucose and galactose; remains a dairy product.                           | Always dairy                    |
| 98  | <b>Lactose-Containing Drinks</b>                     | <b>1 — Mild</b>        | Beverages formulated with lactose for sweetness, browning, or as a carrier.                                                | Always dairy                    |
| 99  | <b>Lactulose</b>                                     | <b>2 — Moderate</b>    | Synthetic disaccharide (galactose + fructose) made from lactose. Used as a laxative and prebiotic.                         | Dairy-derived starting material |
| 100 | <b>Lanolin / Wool Fat</b>                            | <b>4 — Severe</b>      | Wax secreted by the sebaceous glands of sheep, recovered from wool. Used in cosmetics, vitamin D sources, and ointments.   | Animal-derived                  |
| 101 | <b>Lard (if not pure)</b>                            | <b>4 — Severe</b>      | Rendered pork fat. Commercial lard often contains stabilizers, antioxidants, and trace meat solids.                        | Animal-derived                  |
| 102 | <b>Magnesium Lactate</b>                             | <b>2 — Moderate</b>    | Magnesium salt of lactic acid used as a mineral supplement and acidity regulator.                                          | Often dairy-fermented           |
| 103 | <b>Maleates</b>                                      | <b>4 — Severe</b>      | Salts and esters of maleic acid; used as acidulants and intermediates in food and pharmaceutical processing.               | Synthetic additive              |
| 104 | <b>Malted Milk</b>                                   | <b>1 — Mild</b>        | Powder of malted barley, wheat flour, and whole milk used as a beverage flavoring.                                         | Always dairy                    |
| 105 | <b>Marshmallow</b>                                   | <b>3 — Mod. Severe</b> | Aerated confection of sugar, gelatin, and corn syrup; some formulations contain dairy.                                     | Animal-derived gelatin          |
| 106 | <b>Milk</b>                                          | <b>1 — Mild</b>        | Cow's milk - the user's reference baseline for Severity 1 (Mild).                                                          | Always dairy, baseline          |
| 107 | <b>Milk Fat</b>                                      | <b>1 — Mild</b>        | Lipid fraction of milk (label term); when concentrated becomes butter or AMF.                                              | Always dairy                    |
| 108 | <b>Milk Protein / Milk Protein Concentrate (MPC)</b> | <b>1 — Mild</b>        | Co-precipitated casein and whey proteins (>=40% protein for MPC). Used in dairy beverages, cheese, and nutrition products. | Always dairy                    |
| 109 | <b>Milk Solids / Modified Milk Ingredients</b>       | <b>1 — Mild</b>        | Generic label terms covering any non-fluid dairy component (powders, concentrates, isolates).                              | Always dairy                    |

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| 110 | <b>Monoglycerides</b>                    | <b>2 — Moderate</b>    | Fatty acid emulsifiers (mono- and diglycerides) used in baked goods, margarine, and ice cream.                                                                                                                                                                                                                              | Occasionally dairy-derived      |
| 111 | <b>Natural Coloring / Colouring</b>      | <b>4 — Severe</b>      | Pigments from plant, mineral, or animal sources, formulated with stabilizers and ammonium-containing solvents.                                                                                                                                                                                                              | Ammonium incorporated           |
| 112 | <b>Natural Flavoring / Flavouring</b>    | <b>5 — Suppurative</b> | Concentrated flavor extracts from natural sources delivered in solvent and reaction-flavor systems with high ammonium content.                                                                                                                                                                                              | Concentrated ammonium           |
| 113 | <b>Nitrates / Nitrites</b>               | <b>4 — Severe</b>      | Curing salts (sodium/potassium nitrate and nitrite) used in processed meats. Form N-nitroso compounds during cooking.                                                                                                                                                                                                       | Processed meat additive         |
| 114 | <b>Nougat</b>                            | <b>3 — Mod. Severe</b> | Confection made of sugar, honey, whipped egg whites, and often milk solids and butter.                                                                                                                                                                                                                                      | Often dairy                     |
| 115 | <b>OPTA (proprietary brand)</b>          | <b>2 — Moderate</b>    | Brand of proprietary food ingredients developed to enhance texture and healthfulness across dairy, sauces, dressings, meat, and bakery products. Product line includes fat replacers, texturizers, stabilizers for dairy and baked goods, and dressing/sauce systems - typically built on whey or casein protein platforms. | Dairy-protein-based             |
| 116 | <b>Paneer</b>                            | <b>2 — Moderate</b>    | Fresh, non-melting acid-set cheese common in South Asian cuisine.                                                                                                                                                                                                                                                           | Always dairy                    |
| 117 | <b>Pepperoni</b>                         | <b>4 — Severe</b>      | Cured, dried, spiced beef-and-pork sausage with nitrites, paprika, and milk powders.                                                                                                                                                                                                                                        | Processed meat                  |
| 118 | <b>Powdered Milk</b>                     | <b>1 — Mild</b>        | Spray-dried cow's milk in whole or non-fat form.                                                                                                                                                                                                                                                                            | Always dairy                    |
| 119 | <b>Prescription Capsules (Bovine)</b>    | <b>4 — Severe</b>      | Pharmaceutical capsule shells made from bovine gelatin.                                                                                                                                                                                                                                                                     | Bovine-derived                  |
| 120 | <b>Processed Cheese</b>                  | <b>2 — Moderate</b>    | Pasteurized blends of natural cheese with emulsifying salts (often sodium phosphate, citrate) and milk solids.                                                                                                                                                                                                              | Always dairy, processed         |
| 121 | <b>Processed Meats</b>                   | <b>4 — Severe</b>      | Cured, smoked, salted, or fermented meats with binders, nitrates, milk powders, and flavorings.                                                                                                                                                                                                                             | Animal-derived, additive-rich   |
| 122 | <b>Recaldent (CPP-ACP)</b>               | <b>2 — Moderate</b>    | Casein phosphopeptide-amorphous calcium phosphate used in dental products. Direct casein derivative.                                                                                                                                                                                                                        | Always dairy                    |
| 123 | <b>Rennet Casein</b>                     | <b>2 — Moderate</b>    | Casein precipitated by rennet enzymes (rather than acid). Used in imitation cheeses and processed cheese.                                                                                                                                                                                                                   | Always dairy                    |
| 124 | <b>Salami</b>                            | <b>4 — Severe</b>      | Cured fermented sausage with pork, beef, salt, nitrites, spices, and milk powders.                                                                                                                                                                                                                                          | Processed meat                  |
| 125 | <b>Salt (commercial blends)</b>          | <b>2 — Moderate</b>    | Sodium chloride, sometimes blended with anti-caking agents, dextrose carriers, or iodate stabilizers.                                                                                                                                                                                                                       | Usually neutral; some carriers  |
| 126 | <b>SAPP / Sodium Acid Pyrophosphate</b>  | <b>4 — Severe</b>      | Leavening acid and color stabilizer in processed meats, potatoes, and baked goods.                                                                                                                                                                                                                                          | Synthetic additive              |
| 127 | <b>Sherbet</b>                           | <b>3 — Mod. Severe</b> | Frozen dessert with 1-2% milk fat plus fruit juice and sugar; lower fat than ice cream.                                                                                                                                                                                                                                     | Always dairy                    |
| 128 | <b>Simplese (Fat Replacer)</b>           | <b>2 — Moderate</b>    | Microparticulated whey-protein-based fat replacer used in low-fat dairy desserts and dressings.                                                                                                                                                                                                                             | Always dairy                    |
| 129 | <b>Skim Milk</b>                         | <b>1 — Mild</b>        | Cow's milk with butterfat removed (<0.5% fat).                                                                                                                                                                                                                                                                              | Always dairy                    |
| 130 | <b>Soda Pop with Caramel Colouring</b>   | <b>5 — Suppurative</b> | Carbonated soft drinks colored with Class III/IV ammoniated caramel.                                                                                                                                                                                                                                                        | Concentrated ammonium           |
| 131 | <b>Sodium Caseinate</b>                  | <b>2 — Moderate</b>    | Casein neutralized with sodium hydroxide. Used as an emulsifier and protein source in coffee creamers, processed meats, and bakery items.                                                                                                                                                                                   | Always dairy                    |
| 132 | <b>Sodium Stearoyl-2-lactylate (SSL)</b> | <b>4 — Severe</b>      | Dough conditioner and emulsifier made from stearic acid and lactic acid.                                                                                                                                                                                                                                                    | Often dairy-derived lactic acid |
| 133 | <b>Sour Cream</b>                        | <b>2 — Moderate</b>    | Cream fermented with lactic acid bacteria.                                                                                                                                                                                                                                                                                  | Always dairy                    |
| 134 | <b>Sour Milk Solids</b>                  | <b>1 — Mild</b>        | Dehydrated solids from acidified milk, used as flavoring and acidifier in baked goods.                                                                                                                                                                                                                                      | Always dairy                    |
| 135 | <b>Spice</b>                             | <b>5 — Suppurative</b> | Dried aromatic plant material; commercial spice blends are often standardized with ammoniated carriers and may include caramel for color uniformity.                                                                                                                                                                        | Concentrated ammonium           |
| 136 | <b>Spices</b>                            | <b>5 — Suppurative</b> | Plural label form covering proprietary spice blends with stabilizers, anti-caking agents, and ammonia-treated reaction flavors.                                                                                                                                                                                             | Concentrated ammonium           |

|     |                                       |                     |                                                                                                                                               |                                 |
|-----|---------------------------------------|---------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------|
| 137 | <b>Stearates</b>                      | <b>4 — Severe</b>   | Salts and esters of stearic acid (e.g., magnesium stearate, calcium stearate). Used as anti-caking agents, lubricants, and tablet excipients. | Often animal-derived            |
| 138 | <b>Stearic Acid</b>                   | <b>4 — Severe</b>   | Saturated fatty acid commonly rendered from beef tallow or pork fat. Used in confectionery, cosmetics, and pharmaceuticals.                   | Often animal-derived            |
| 139 | <b>Suet</b>                           | <b>4 — Severe</b>   | Hard fat surrounding beef or mutton kidneys, used in puddings, pastries, and bird feed.                                                       | Animal-derived                  |
| 140 | <b>Sulfites / Sulphites</b>           | <b>4 — Severe</b>   | Sulfur dioxide and sulfite salts used as preservatives, antimicrobials, and antioxidants in wine, dried fruit, and processed foods.           | Synthetic preservative          |
| 141 | <b>Tagatose</b>                       | <b>2 — Moderate</b> | Low-calorie sweetener (a ketose isomer of fructose) manufactured from lactose via galactose isomerization.                                    | Dairy-derived starting material |
| 142 | <b>Tallow</b>                         | <b>4 — Severe</b>   | Rendered beef or mutton fat, used in frying, soap, and shortening blends.                                                                     | Animal-derived                  |
| 143 | <b>Whey</b>                           | <b>2 — Moderate</b> | Liquid serum remaining after milk is curdled and strained. Sweet whey from rennet cheese; acid whey from cottage cheese/yogurt.               | Always dairy                    |
| 144 | <b>Whey Protein Concentrate (WPC)</b> | <b>2 — Moderate</b> | Whey concentrated to 25-89% protein. Used in sports nutrition, bars, and bakery.                                                              | Always dairy                    |
| 145 | <b>Whey Protein Isolate (WPI)</b>     | <b>2 — Moderate</b> | Whey concentrated to >=90% protein with most fat and lactose removed.                                                                         | Always dairy                    |
| 146 | <b>Whiskey</b>                        | <b>4 — Severe</b>   | Distilled spirit from fermented grain mash. May use ammonium-containing yeast nutrients during fermentation.                                  | Possible ammonium in production |
| 147 | <b>Wine (Red and White)</b>           | <b>4 — Severe</b>   | Fermented grape beverage. Often clarified with casein or egg whites; commonly contains added sulfites and may include caramel for adjustment. | Possible dairy fining; sulfites |
| 148 | <b>Yogurt</b>                         | <b>2 — Moderate</b> | Milk fermented with <i>Lactobacillus bulgaricus</i> and <i>Streptococcus thermophilus</i> .                                                   | Always dairy                    |

## 4. Application Notes

Consumer avoidance: Acnerase cross-references all identifiable ingredient terms and their 150+ synonyms in real-time product scanning to generate a risk-stratification output.

Elimination protocol: Severity 4–5 ingredients are the highest-priority first targets, producing the greatest reduction in inflammatory burden per unit of dietary change.

Research use: Standard food frequency questionnaires do not capture these ingredient categories and will substantially underestimate relevant antigenic burden in susceptible populations. This index provides the specific categories required for accurate dietary Neu5Gc and ammonium catalyst assessment.

## 5. Observational Foundation

The 148 ingredients were identified through 23 years of deliberate, systematic self-provocation — each ingredient category consumed separately, inflammatory responses observed, lesion characteristics documented, and observations repeated to confirm reproducibility. The N-of-1 limitation is real and acknowledged throughout the manuscript series. So is the uniqueness of the dataset. This index is a starting point for formal investigation, not a substitute for it.